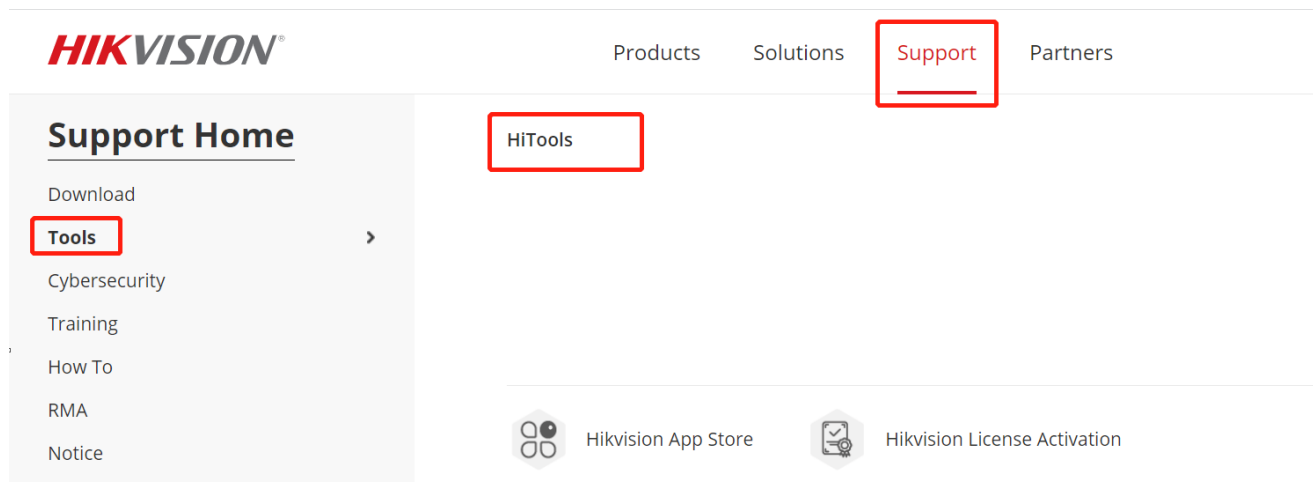


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|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 1 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |

## How To Use Thermal Design Tool

### ONLINE ADDRESS

There are two tool paths, one is to log in through Hikvision's official website, Support-Tools-Hitools, search for Thermal design tool to find, click on the home page in the upper right corner to select different languages;



The other path is directly accessed through the URL address below:

<https://tdt.hikmicrotech.com/index#/lensSelection-thermal>

### IMPORT PICTURE MAP

#### STEP 1: Get the image from google earth

For the completed project site, we can choose to import the project picture, search for the location of the specific site through Google, use the online ranging function, select two points to get the distance and save the screenshot

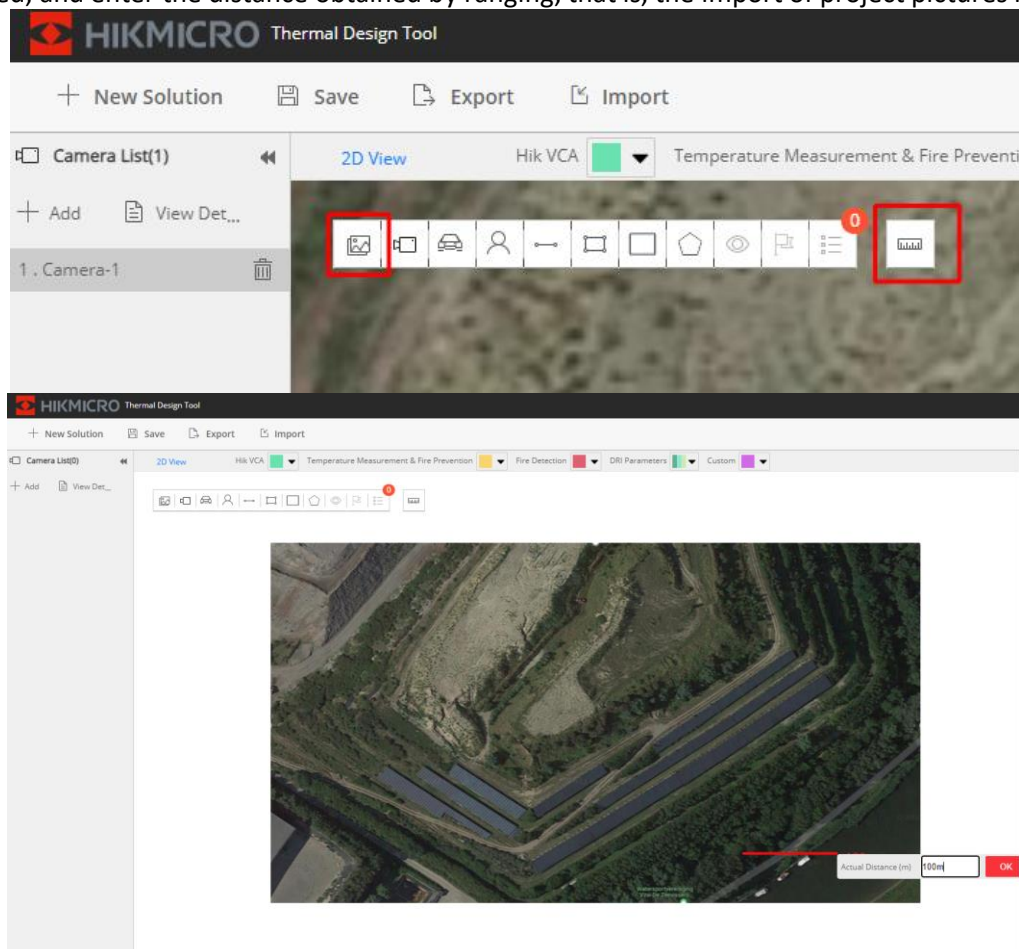
- Google to get the right site
- Draw a line in two points, get the real distance
- Save the screenshot

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|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 2 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |



## STEP 2: Import picture and click edit scale

Use the image import function to import screenshots, and then use **scale** to set the scale bar, corresponding to the two points just selected, and enter the distance obtained by ranging, that is, the import of project pictures is completed;



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|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 3 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |

## VCA CONFIGURATION

### STEP 1: Choose camera

Next we can see that there are various intelligent parameters to choose, the default is the VCA function, then we take VCA as an example, click on the camera to select the specific model, and the selection interface has a variety of filtering methods for us to choose:

**Focal length:** Customers who are very familiar with Hikvision products can enter the specific lens millimeter to select the model;

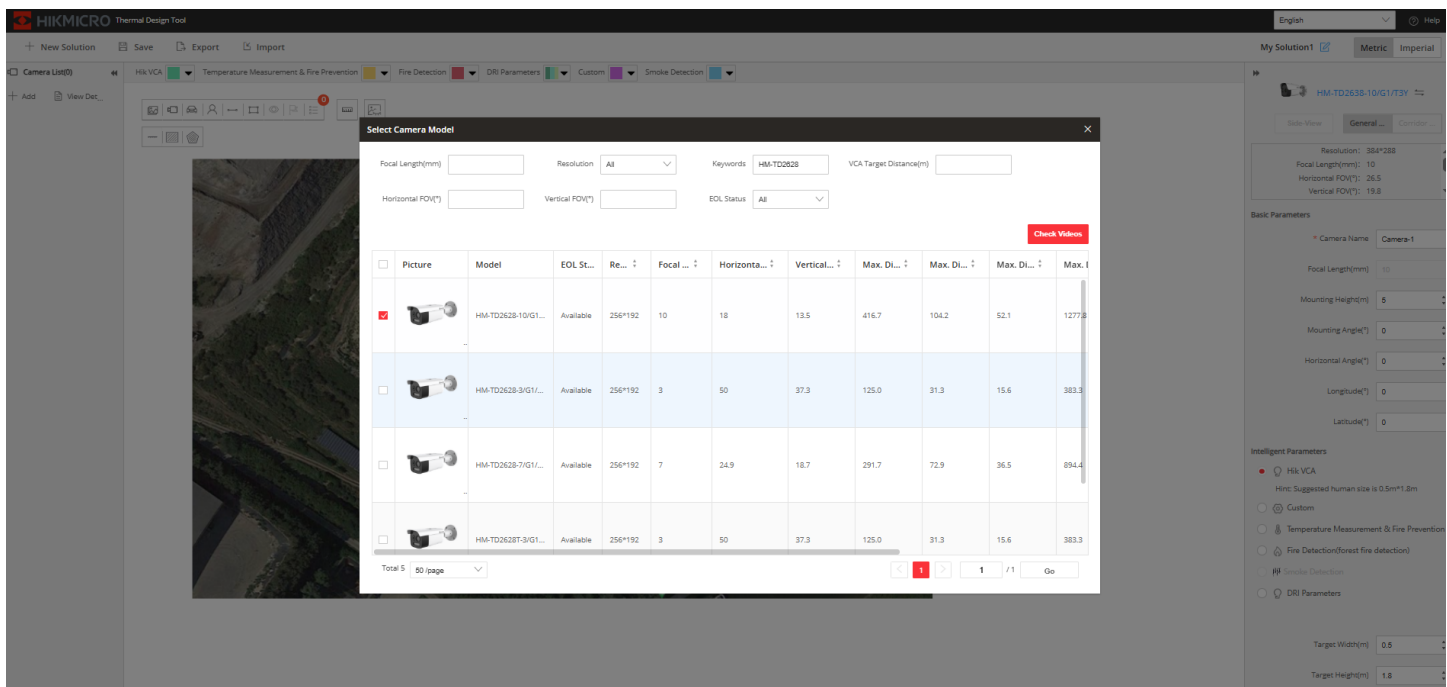
**Resolution:** You can choose low-resolution heatpro or high-end 640 products according to your budget;

**Keywords:** Customers familiar with Hikvision products can enter some product model names to quickly locate the required model range;

**VCA target distance:** If you are not familiar with Hikvision products, you can enter the target distance to be detected, and filter the model range that can meet the detection distance;

**FOV field of view:** For customers with field of view needs, you can also do certain screening here.

**EOL Status:** When selecting cameras, can filter based on product status (EOL/Available)



During perimeter camera model selection, customers may not have a clear understanding of the imaging performance of devices with different fields of view (FOV) and focal lengths. This function supports viewing video effects from fixed cameras and allows simultaneous playback of videos from two device models for performance comparison.

### Currently supported models:

DS-2TD1228-xx, DS-2TD2628-xx, DS-2TXS2628-xx, HM-TD1228-xx, HM-TD2628-xx, HM-TD2638-xx, HM-TD2668-xx.

|                |                                |                 |      |              |           |
|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 4 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |

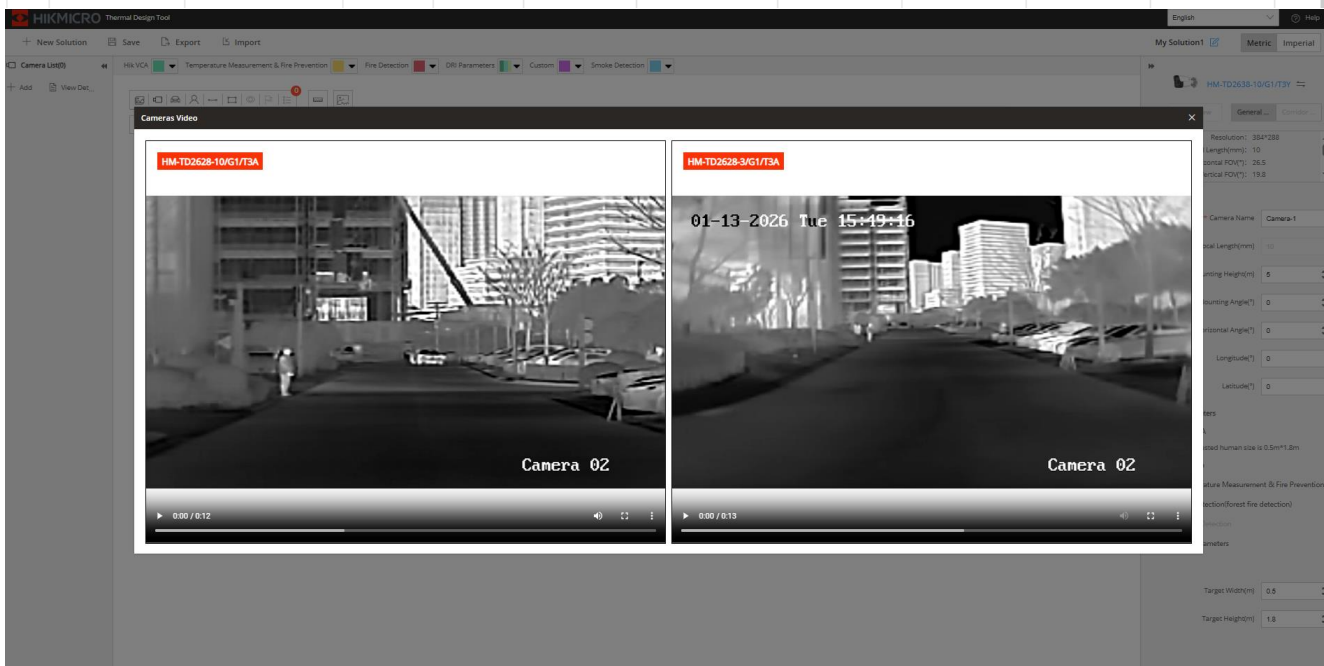
## Select Camera Model

Focal Length(mm)  Resolution  Keywords  VCA Target Distance(m)

Horizontal FOV(°)  Vertical FOV(°)  EOL Status

[Check Videos](#)

| <input type="checkbox"/>            | Picture                                                                           | Model               | EOL St... | Re... ↑ | Focal ... ↑ | Horizonta... ↑ | Vertical... ↑ | Max. Di... ↑ | Max. Di... ↑ | Max. Di... ↑ | Max. I |
|-------------------------------------|-----------------------------------------------------------------------------------|---------------------|-----------|---------|-------------|----------------|---------------|--------------|--------------|--------------|--------|
| <input checked="" type="checkbox"/> |  | HM-TD2628-10/G1/... | Available | 256*192 | 10          | 18             | 13.5          | 416.7        | 104.2        | 52.1         | 1277.8 |
| <input checked="" type="checkbox"/> |  | HM-TD2628-3/G1/...  | Available | 256*192 | 3           | 50             | 37.3          | 125.0        | 31.3         | 15.6         | 383.3  |

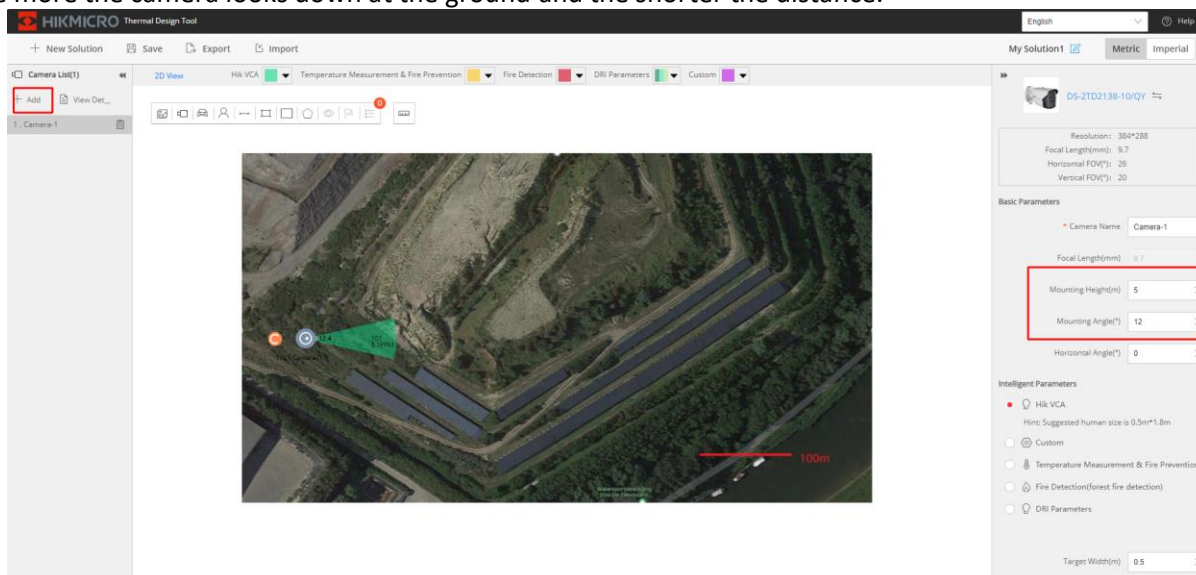


## STEP 2: Add camera into the map and move it to the right site

After selecting the camera, add it in the left camera bar, and place the camera in a suitable position, the mouse selects the yellow button to adjust the horizontal angle of the camera, of which the green part is the VCA coverage area, the gray part is out of sight area, and there is a corresponding distance display on it; At the same time, we can also adjust the mounting height and vertical angle of the camera in the basic parameter bar on the right. The default vertical angle under the VCA function is to ensure the longest distance angle, we can adjust it as needed, if the actual installation can only be a certain height, then input this height into TDT, and then adjust the vertical angle of the camera to find the

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|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 5 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |

most suitable combination. The larger the installation height, the larger the out of sight zone. The larger the vertical angle, the more the camera looks down at the ground and the shorter the distance.

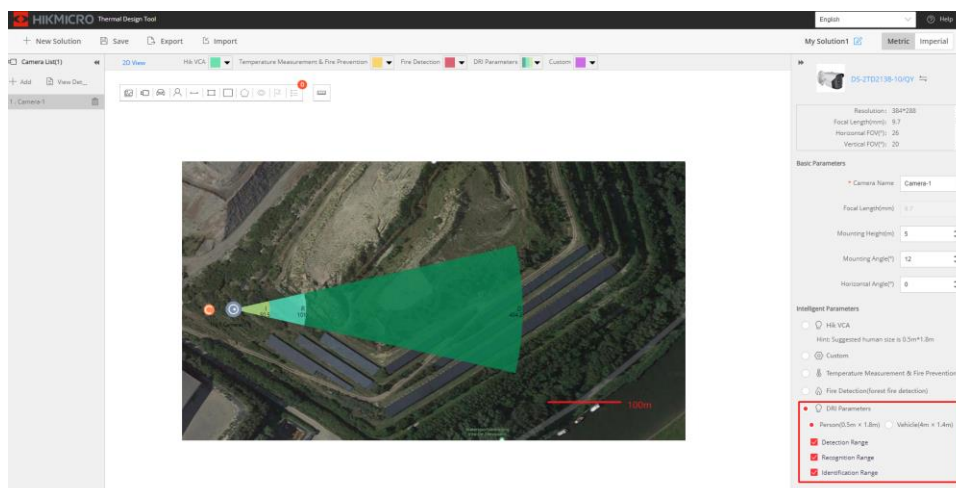


If you install the camera and don't just use it for VCA rules, but also for different functions such as temperature measurement and fire protection, you can choose different intelligent parameters. In addition, we offer DRI Parameters. You can see that the different colors represent Detection range, Recognition range, and Identification range. Different DRI range have different meanings.

**Detection Range:** In order to distinguish an object from the background, the image must be covered by 1.5 or more pixels.

**Recognition Range:** In order to classify the object (animal, human, vehicle, etc.), the image must have at least 6 pixels across its critical dimension.

**Identification Range:** In order to identify the object and describe it in details, the critical dimension must have at least 12 pixels across

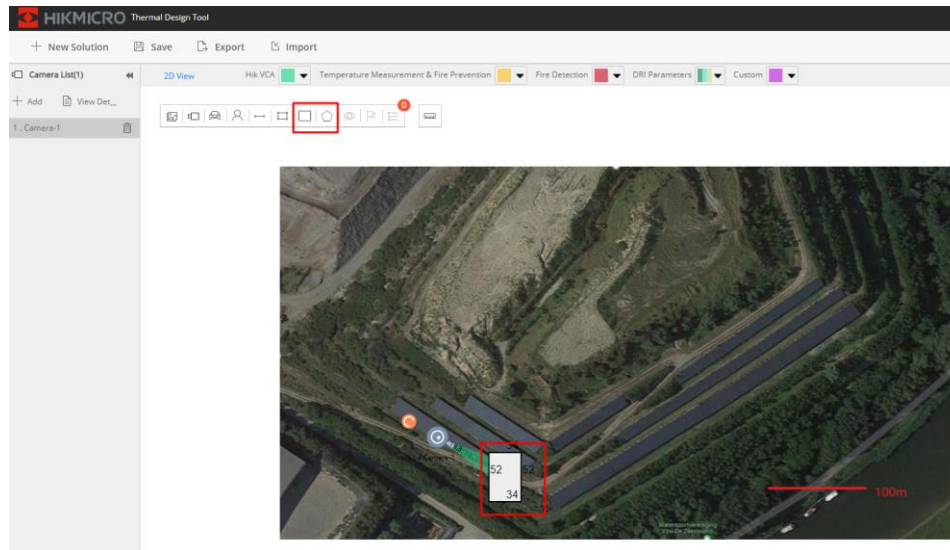




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|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 6 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |

### STEP 3: Draw obstacle area

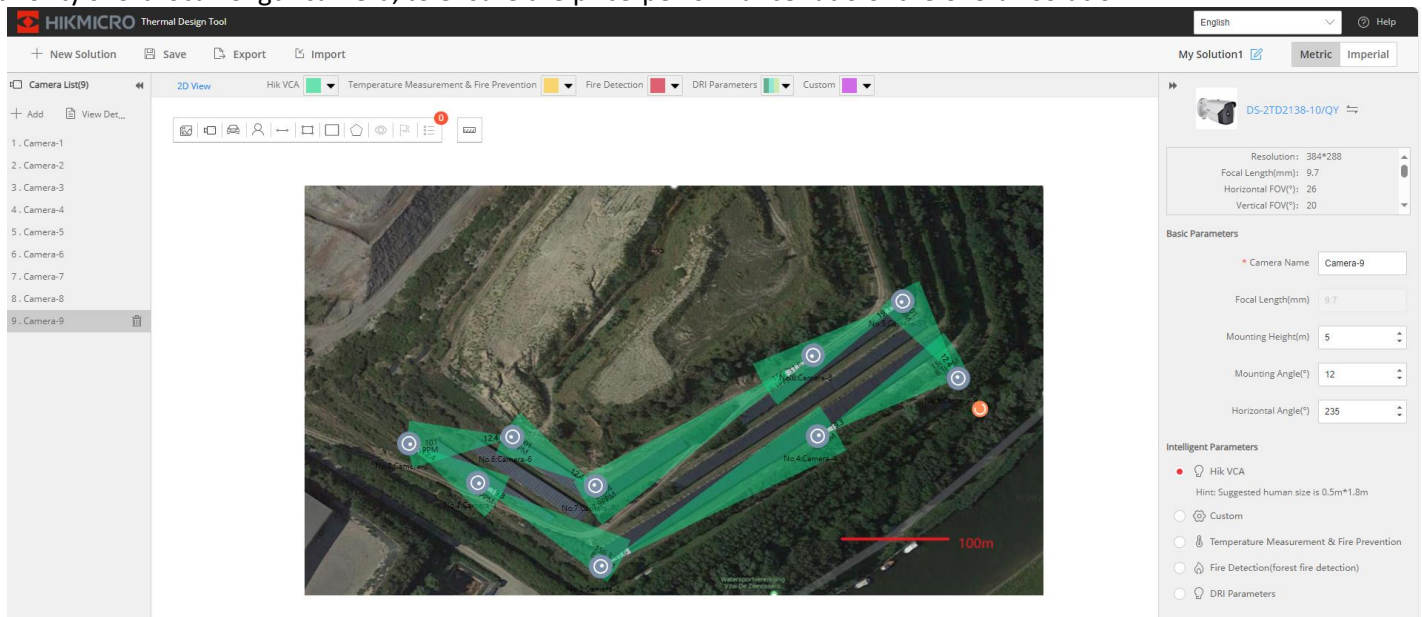
If there are obstacles in the field that will affect the camera's field of view, You can click on rectangle and polygonal in the toolbar to draw the obstacle area to simulate the real situation.



### STEP 4: Continue to add more cameras until covering whole borders

The next step is to add more cameras until the entire perimeter is completely covered, the camera position is preferably connected head to end, and use the previous camera to cover the out of sight zone of the next camera to ensure that there is no uncovered area;

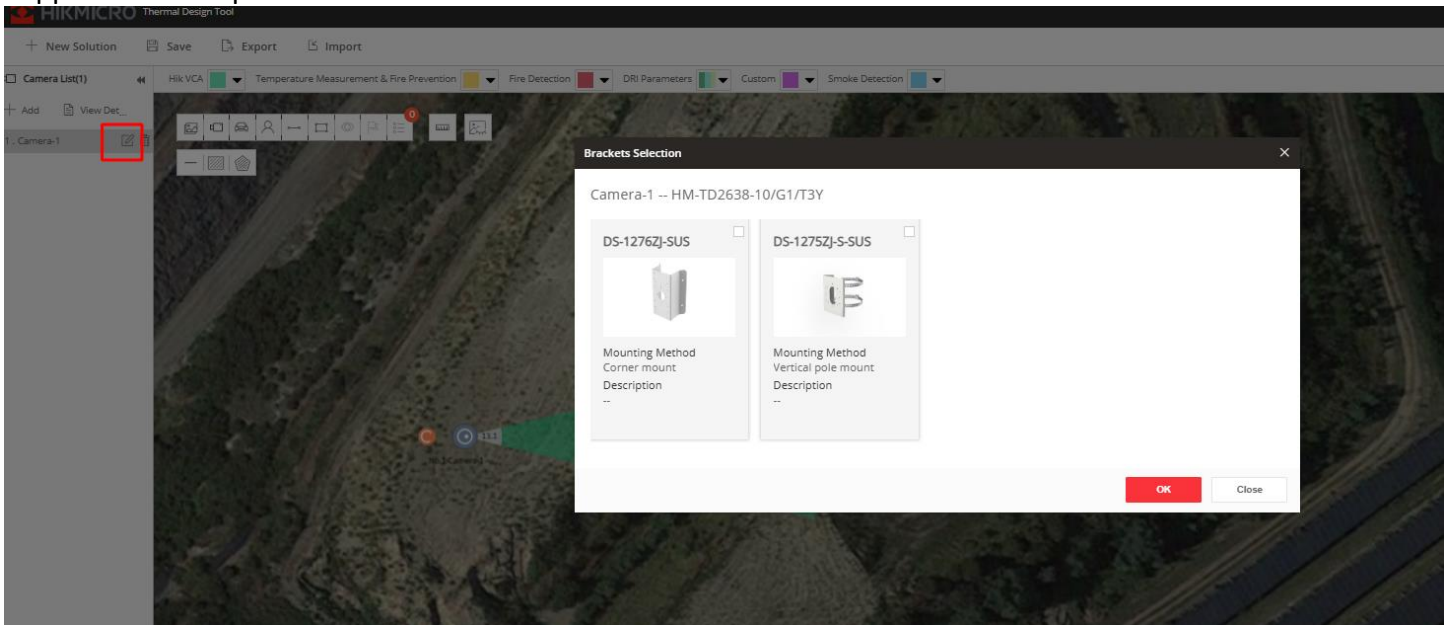
When selecting the camera model, choose according to the long range priority long focal length camera, short distance priority short focal length camera, to ensure the price-performance ratio of the overall solution



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|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 7 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |

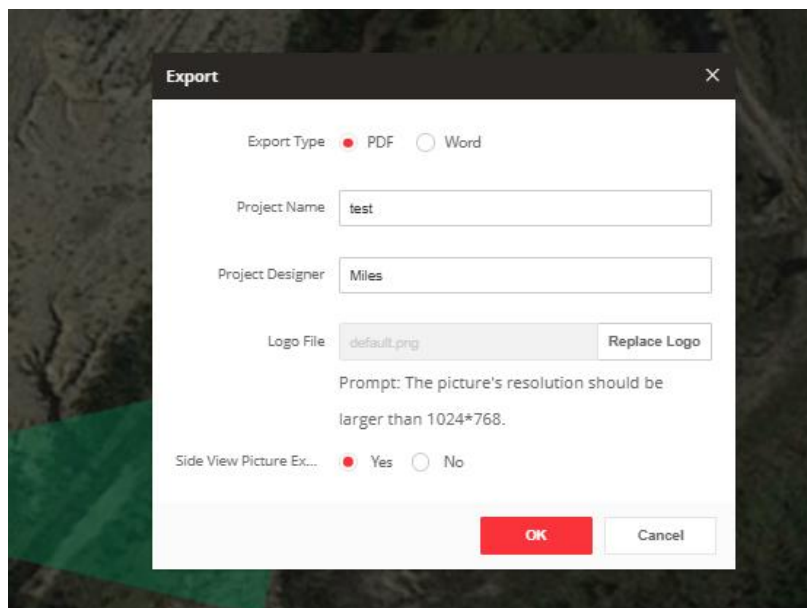
**STEP 5: After the camera and its position are determined, the corresponding bracket and accessories can be selected.**

Supports to be exported in the solution



**STEP 6: Export with PDF report**

After all cameras are added, you can directly export PDF/Word reports, covering project pictures, detailed point information, etc.

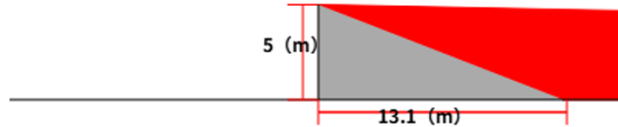


When exporting, the camera side view can be selected, so the bottom of the report will show the camera blind area and installation height& angle in more detail.

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|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 8 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |

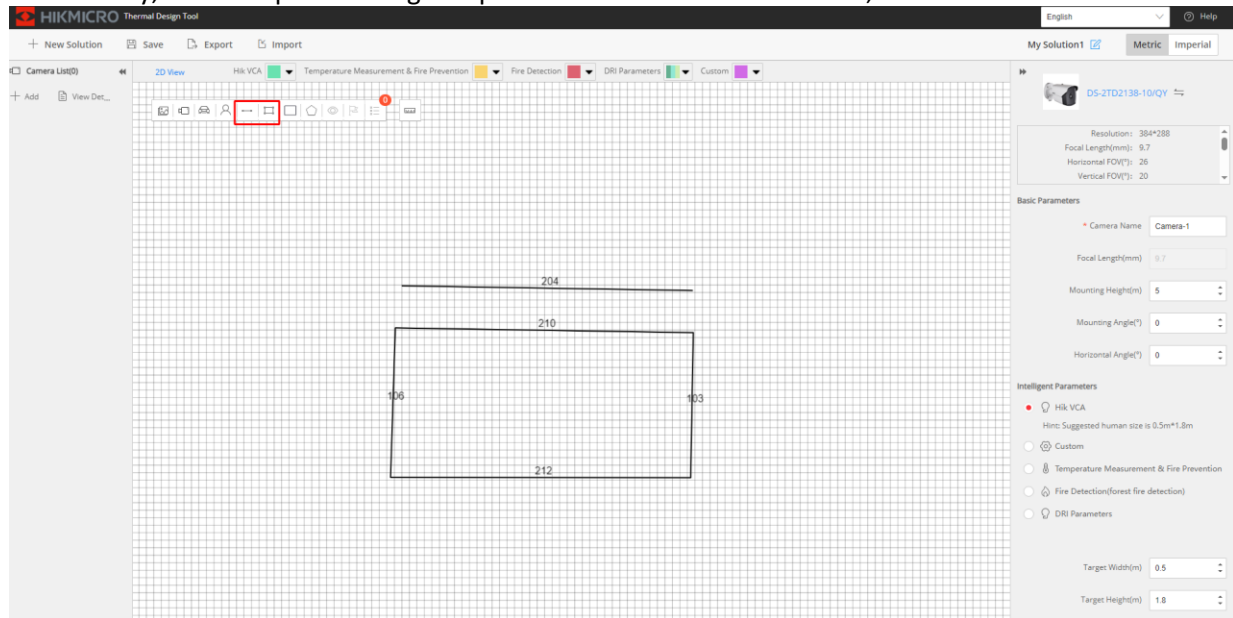
Side-View

Camera- I -HM-TD2638- I O/G I /T3Y



## STEP 7: Project Without Map

If the project is still under construction and there is no specific site picture, it can be directly replaced by a line segment or polygon boundary, and the specific design steps are consistent with the above;

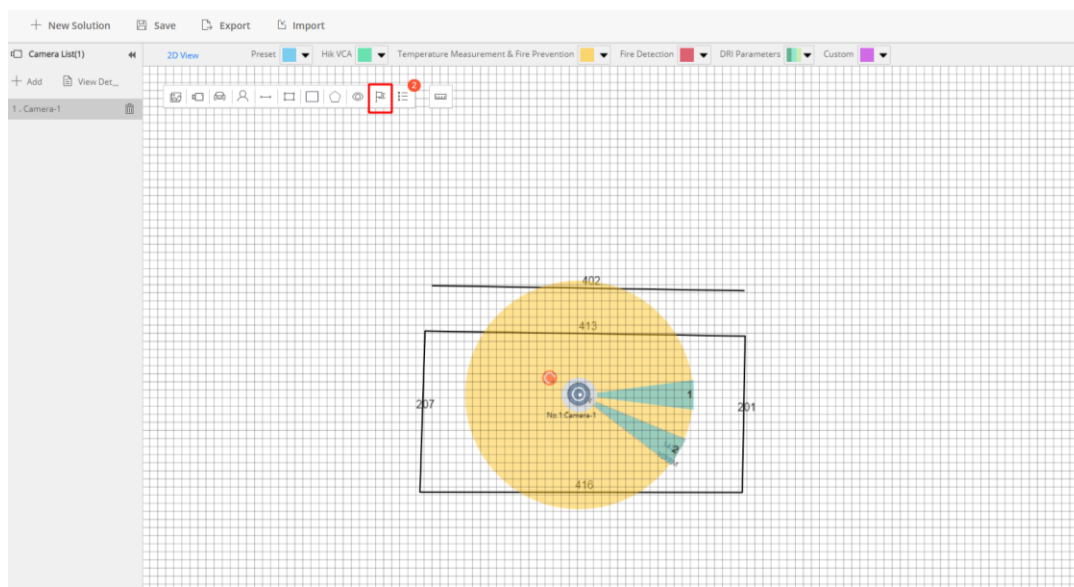


## STEP 8: Preset & PTZ VIEW

We can also add **PTZ** into solution design, we can add **PTZ** preset points, the same rotation yellow button can adjust the viewing angle; Each preset point is independent and can be deleted in the upper toolbar;



|                |                                |                 |      |              |           |
|----------------|--------------------------------|-----------------|------|--------------|-----------|
| <b>Product</b> | Thermal Design Tool            | <b>Version:</b> | V2.0 | <b>Date:</b> | 3/18/2026 |
| <b>Contact</b> | qilongqi@hikvision.com.cn      |                 |      | <b>Page:</b> | 9 of 9    |
| <b>Title</b>   | How to Use Thermal Design Tool |                 |      |              |           |



If you only want to see the field of view of the preset point, we can also hide the field of view of the PTZ except the preset point in the upper toolbar, so that it appears clearer in the export report;

